Alternative Coins (Altcoins)

Chapter 8 — Mastering Blockchain, 2nd Edition

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Blockchain | Cryptocurrency | Altcoins | Privacy | Consensus

Learning Objectives

By the end of this session, you will be able to:

- Define altcoins and distinguish them from alternative blockchains
- Explain methods for initial coin distribution (Proof of Burn, Proof of Ownership, sidechains)
- Compare alternative consensus mechanisms: PoS, PoA, Proof of Coinage, Proof of Deposit
- Describe difficulty adjustment algorithms and their role in altcoin stability
- Identify key limitations of Bitcoin that motivated altcoin development
- Analyze notable altcoins: Namecoin, Litecoin, Primecoin, and Zcash
- Explain what ICOs and ERC20 tokens are and how they function

Chapter Overview: The Altcoin Landscape

Timeline: Bitcoin (2009) → Namecoin (2011, first fork) → Altcoin boom (2013–2014)

Two categories of Bitcoin alternatives:

- Altcoin — primary goal is a new virtual currency
- Alternative Blockchain — new decentralized platform (Chapter 16)

Why do altcoins emerge?

- Address Bitcoin limitations: privacy, speed, energy waste
- Introduce new consensus mechanisms or PoW algorithms
- Provide specialized services (e.g., naming, asset tokenization)
- Experiment with new economic models (e.g., unlimited coin supply)

As of 2018: hundreds of altcoins exist — many failed, a few succeeded through genuine innovation

What Is an Altcoin?

Definition & Creation

Definition

An altcoin is any cryptocurrency whose primary purpose is to introduce a new virtual currency — distinct from Bitcoin.

Three ways to create an altcoin:

- Fork of Bitcoin source code (most common)
- Fork of another altcoin's code
- Written entirely from scratch (e.g., NEM)

- Most rely on Bitcoin's proven security model as a starting point

Hard Forks & Community

Hard Fork = New Altcoin

A permanent chain split creates two separate chains — and two coins.

Example:

- Ethereum (ETH) hard fork → Ethereum Classic (ETC)

Key challenge:

- Altcoins must attract miners, users, and investors
- Value comes from network effect and community acceptance
- Without adoption, an altcoin has no value

Launching an Altcoin: Initial Coin Distribution

Altcoins must give early users a reason to join. Four distribution methods:

1. New Blockchain	Allocate coins to initial miners. Now unpopular due to pump-and-dump scams where initial miners profit then disappear.
2. Proof of Burn (PoB)	Permanently destroy bitcoins to claim altcoins in proportion — a one-way peg. Example: Counterparty (XCP) created via PoB.
3. Proof of Ownership	Prove you hold bitcoins (without destroying them) to claim altcoins. Achieved via merged mining or pegged sidechains.
4. Pegged Sidechains	Separate blockchain linked to Bitcoin; bitcoins can move in and out (two-way peg). Altcoins can also be transferred back.

The Altcoin Market: Top 10 Cryptocurrencies (March 2018)

Coin	Market Cap	Price (USD)
Bitcoin	**$151.4B**	**$8,951.83**
Ethereum	$68.5B	$697.94
Ripple	$31.3B	$0.80
Bitcoin Cash	$17.2B	$1,010.08
Litecoin	$10.0B	$179.11
NEO	$5.6B	$86.74
Cardano	$5.5B	$0.21
Stellar	$5.4B	$0.29
EOS	$4.3B	$6.04
Monero	$4.2B	$266.40

Source: coinmarketcap.com (March 2018)

Alternatives to Proof of Work (PoW)

Bitcoin's PoW issues: energy waste, ASIC centralization — alternatives proposed:

ASIC-Resistant PoW	Memory-hard puzzles; building ASICs requires large memory → expensive hardware. Scrypt (Litecoin), X11 (Dash). Note: both eventually got ASICs.
Proof of Useful Work	PoW that also performs real scientific computation. Primecoin finds prime number chains (Cunningham chains) of genuine scientific value.
Proof of Storage	Miners store large pseudo-random data subsets. Provides useful distributed archival storage. Introduced by Microsoft Research.
Proof of Stake (PoS)	"Virtual mining" — mining rights proportional to coin ownership, not compute power. No energy-intensive hashing. First used in Peercoin (2012).

Bitcoin Energy Consumption: The PoW Problem

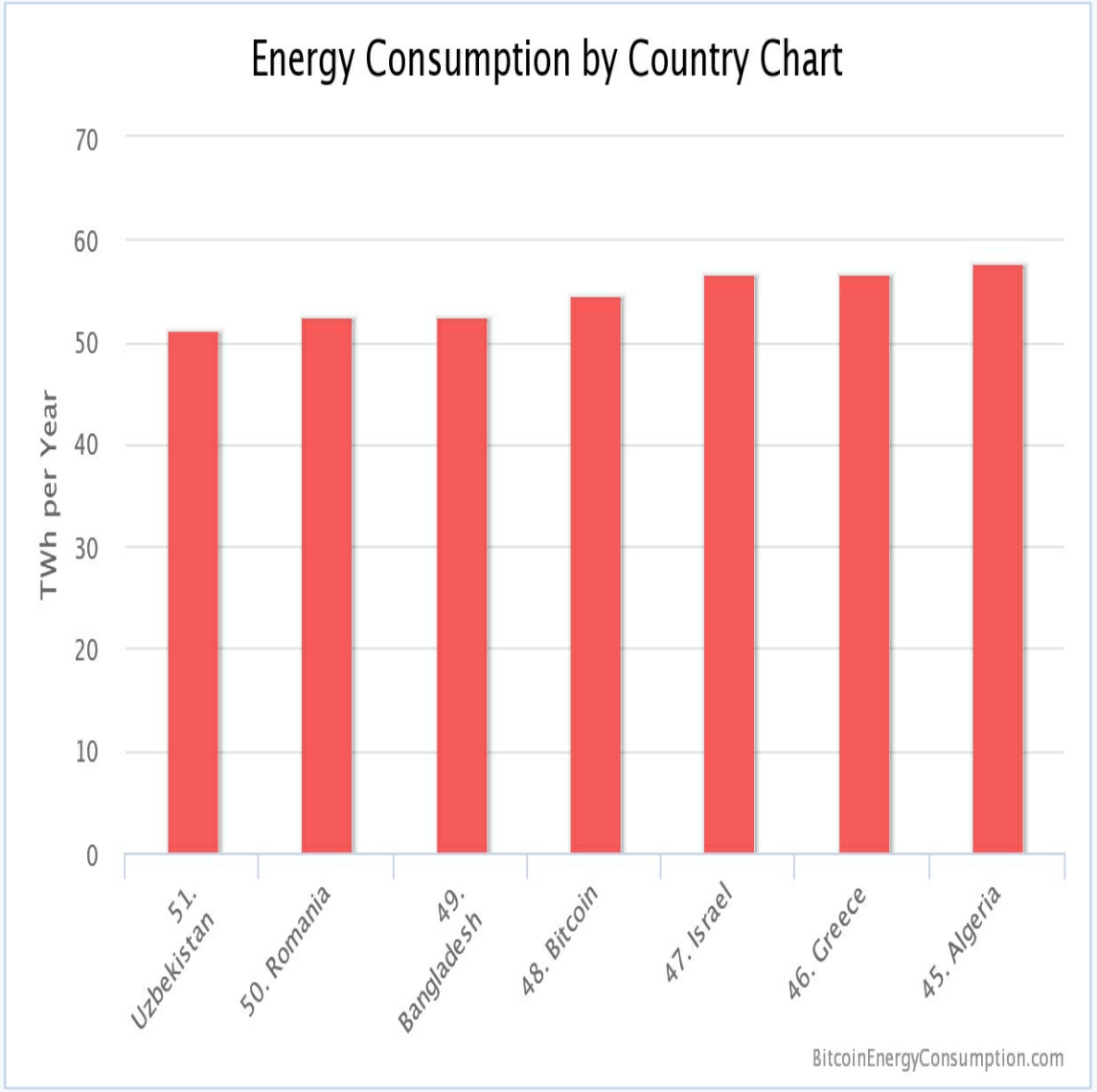
Energy Facts (2018)

- 54.69 terawatt-hour (TWh)/year — exceeds Bangladesh's national consumption
- ~387 kg CO_2 per transaction
- Projected 125 TWh/year by end of 2018

This drove search for alternatives:

- Proof of Stake — no computation needed
- Proof of Useful Work — computation serves science
- Proof of Storage — useful distributed storage

Source: digiconomist.net



Proof of Stake: Variants

Proof of Coinage	Age of coin (time since last spent) determines mining ease. Coinage resets on mining. Rewards holding — implemented in Peercoin combined with PoW.
Proof of Deposit (PoD)	Newly minted coins are locked/frozen for a set number of blocks. Mining at the cost of temporarily freezing coins. A type of PoS.
Proof of Activity (PoA)	Hybrid PoW + PoS: blocks start with PoW, then 3 random stakeholders must digitally sign them. Issue: "nothing at stake" — PoS has no cost so miners can try multiple forks simultaneously.
Nonoutsourceable Puzzles	Miners claim rewards directly rather than through a pool operator. Discourages mining pool formation by creating inherent mistrust between anonymous miners.

Difficulty Adjustment Algorithms

Problem: Bitcoin adjusts difficulty every 2,016 blocks (~2 weeks). Pool hopping exploits slow adjustment. These algorithms solve that.

Algorithm	Key Innovation	Used In
Bitcoin Default	Adjusts every 2,016 blocks (~2 weeks)	Bitcoin
Kimoto Gravity Well (KGW)	Adjusts every block; adaptive response to rapid hash rate spikes	Megacoin, others
Dark Gravity Wave (DGW)	Fixes KGW time-warp attacks; uses exponential moving averages for smoother readjustment	Dash, Bitcoin SegWit2X
DigiShield	Fixed-window readjustment; very fast recovery from hash rate changes	Zcash (v3.0)
MIDAS	Multi-Interval Difficulty Adjustment; rapid response, blocks time-warp attacks	Various altcoins

Bitcoin Privacy Limitations & CoinJoin

Bitcoin is pseudonymous, NOT anonymous

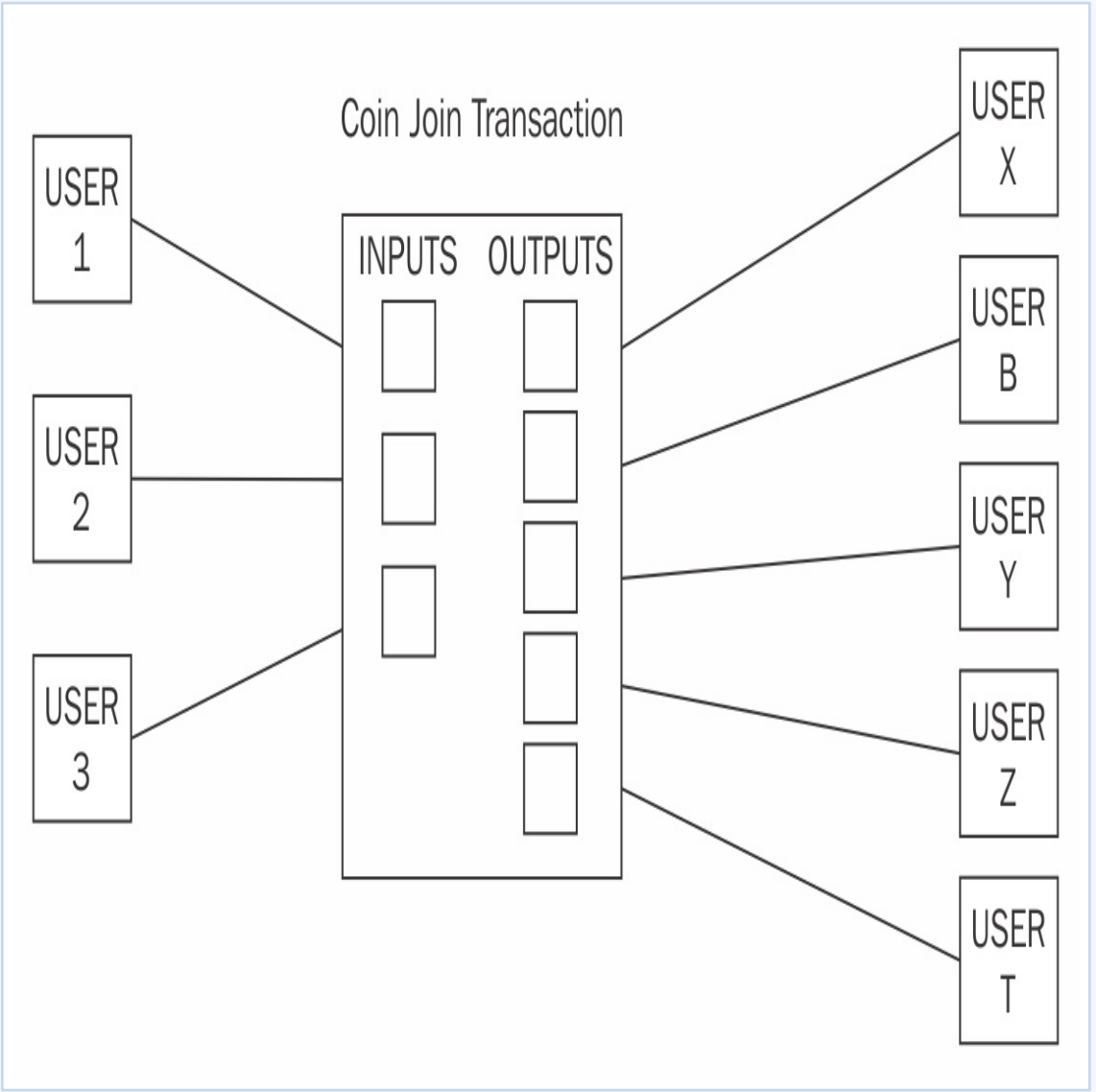
- Public blockchain + traffic analysis = traceable transactions
- IP addresses can be linked to transactions

Three approaches to privacy:

- **Mixing protocols (e.g., CoinJoin)**
  Two transactions combine — inputs/outputs become indistinguishable

- **Third-party mixers (e.g., CoinShuffle, Darksend)**
  Decentralized mixing without a trusted intermediary

- **Inherent anonymity (e.g., Zcash, Monero)**
  Privacy built into coin design using advanced cryptography



Extended Protocols: Colored Coins & Counterparty

Colored Coins

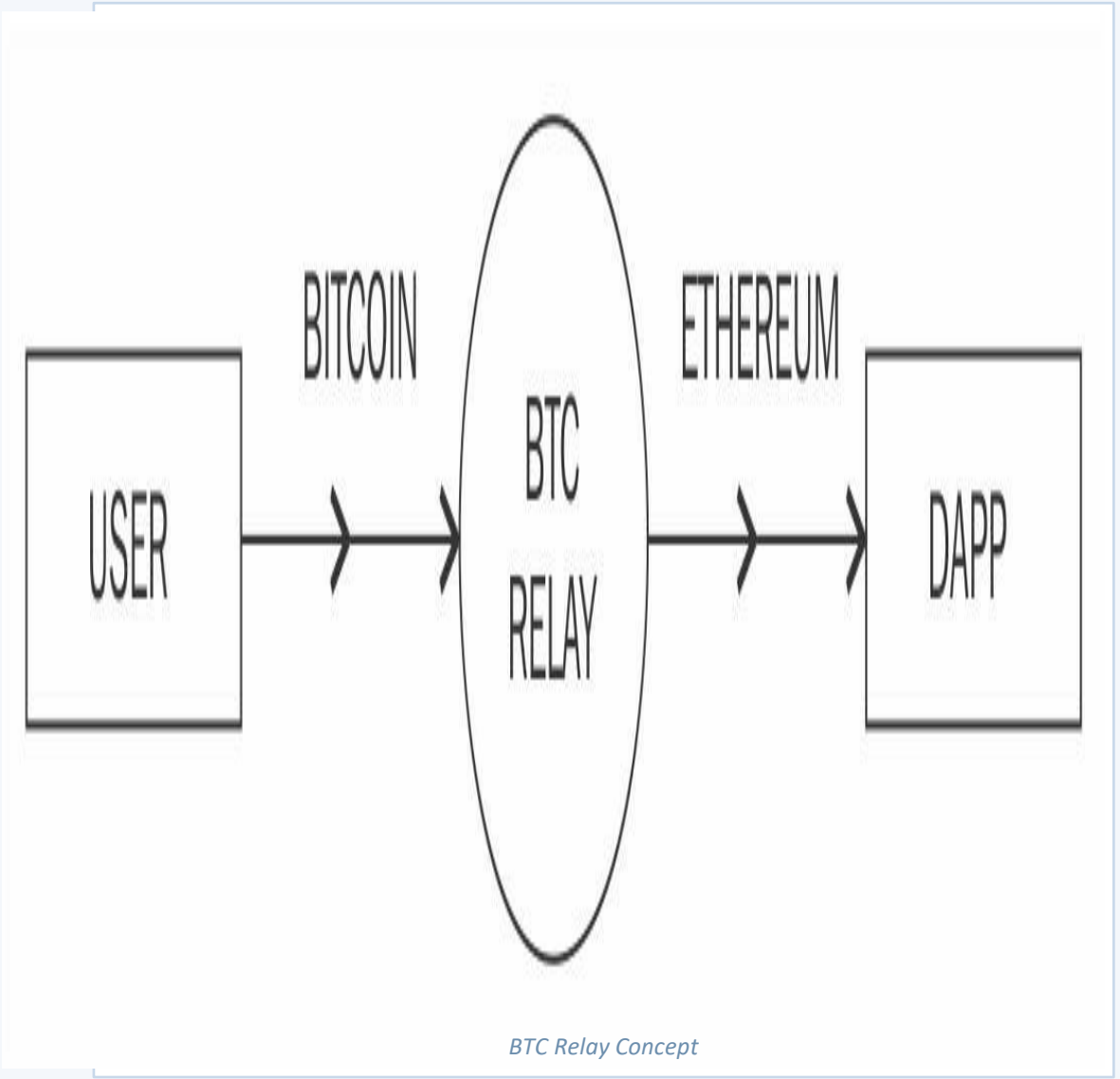
- Attach metadata to specific Bitcoins → represent digital assets
- Uses OP_RETURN opcode in Bitcoin transactions
- Use cases: shares, bonds, certificates, voting rights

Counterparty (XCP)

- Platform for custom tokens on top of Bitcoin
- Uses embedded consensus via OP_RETURN
- XCP coins created via Proof of Burn

BTC Relay (diagram right):

- Ethereum contract that reads Bitcoin block headers
- Enables cross-chain smart contract interaction



BTC Relay Concept

Namecoin — The First Bitcoin Fork (2011)

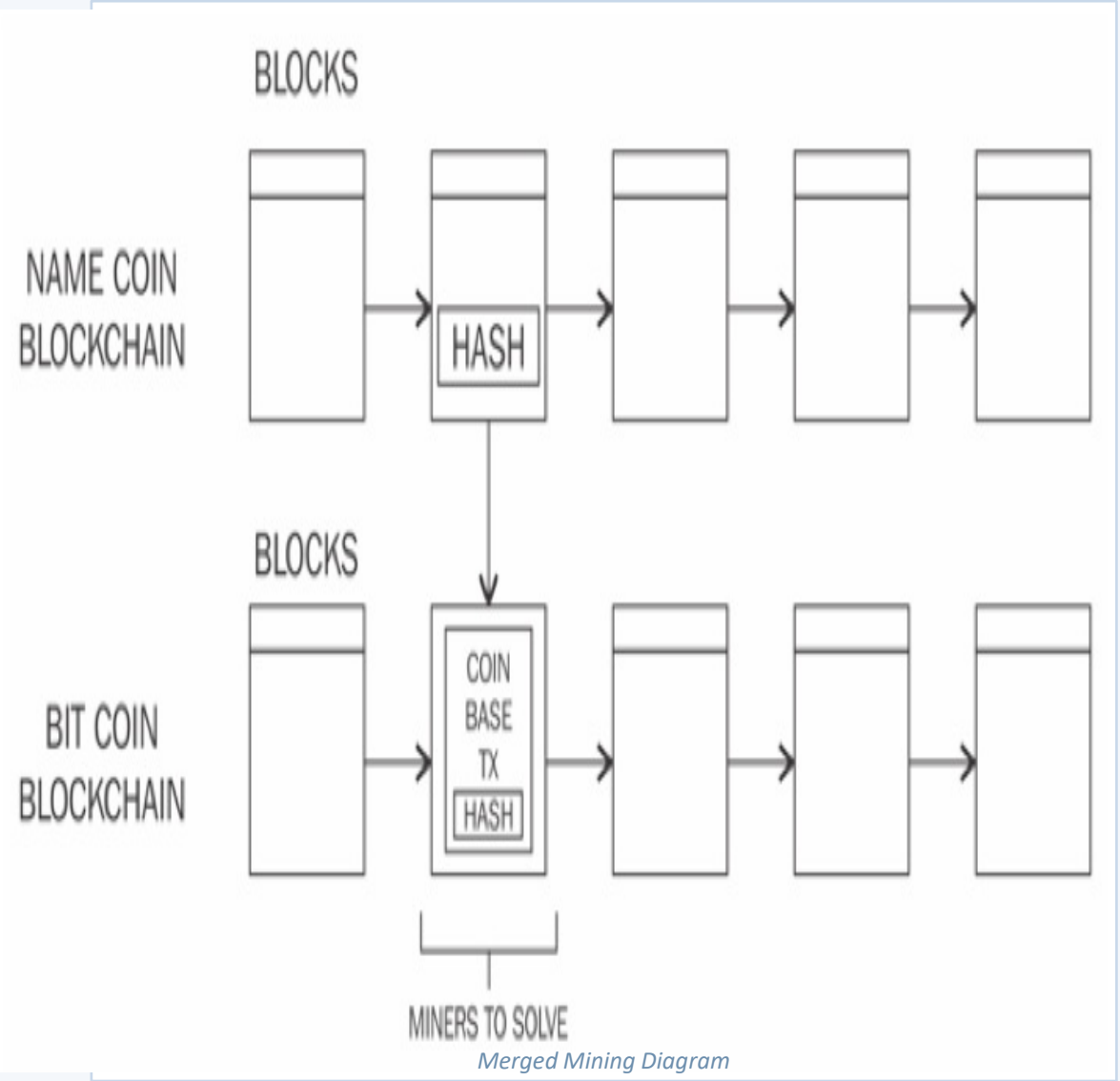
Purpose: Decentralized Naming (not currency)

- First fork of Bitcoin source code
- First solution to Zooko's Triangle
- Introduces .bit top-level domain names
- Decentralized TLS certificate validation

Key Innovation: Merged Mining

- Bitcoin miners also mine Namecoin simultaneously
- Same computational work secures both chains
- Hash of Namecoin block embedded in Bitcoin coinbase tx

Three services: name/key storage, value attachment, NMC currency



Merged Mining Diagram

Litecoin — Faster Transactions with Scrypt PoW

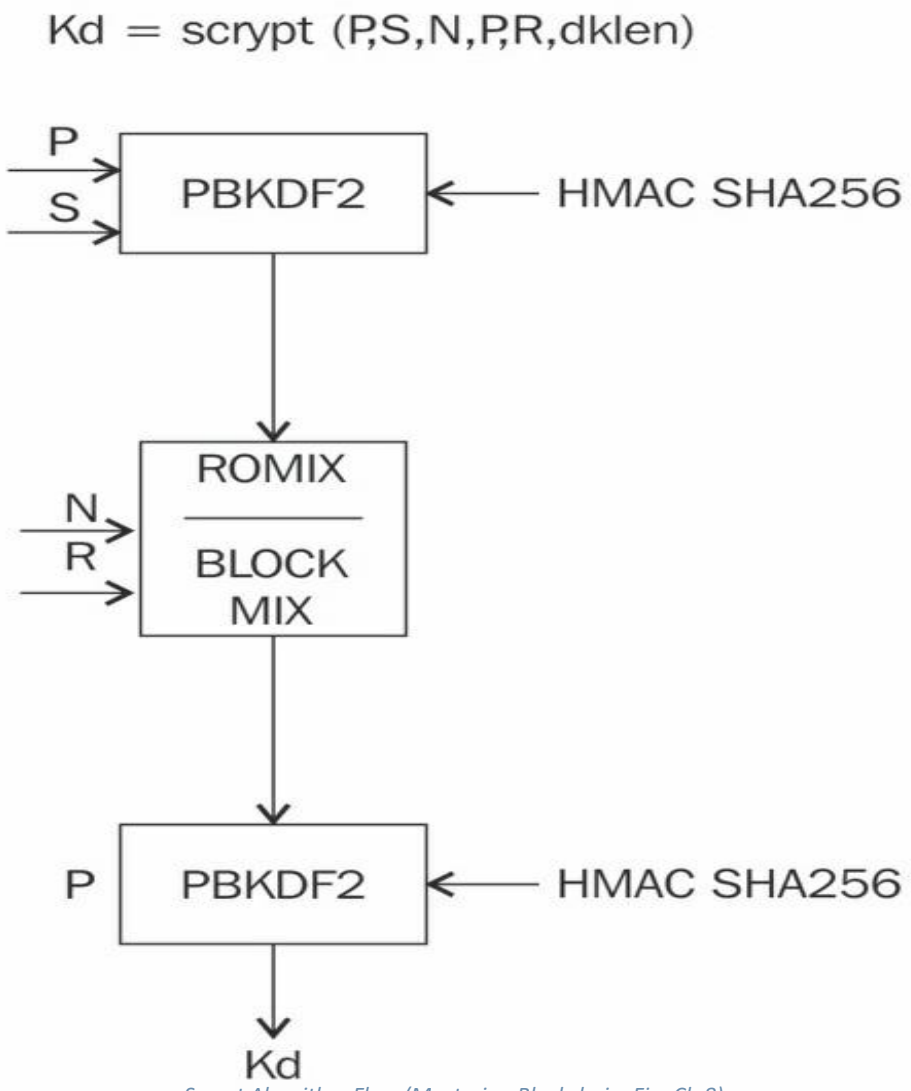
Released 2011 — Fork of Bitcoin

- Block time: 2.5 min vs. Bitcoin's 10 min (4× faster)
- Total supply: 84 million LTC (4× Bitcoin)
- Difficulty readjusted every ~3.5 days

Scrypt PoW:

- Sequential memory-hard function
- Time-Memory Trade-Off (TMTO): More memory = faster; less memory = slower computation
- Originally ASIC-resistant — Scrypt ASICs now exist
- Parameters: N=1024, R=1, P=1, 256-bit output

Mining: CPU → GPU → ASIC (Scrypt Miner Wolf, 2 Gh/s)



Scrypt Algorithm Flow (Mastering Blockchain, Fig. Ch.8)

Primecoin — Proof of Useful Work

First Cryptocurrency with Proof of Useful Work

PoW = Finding Prime Number Chains:

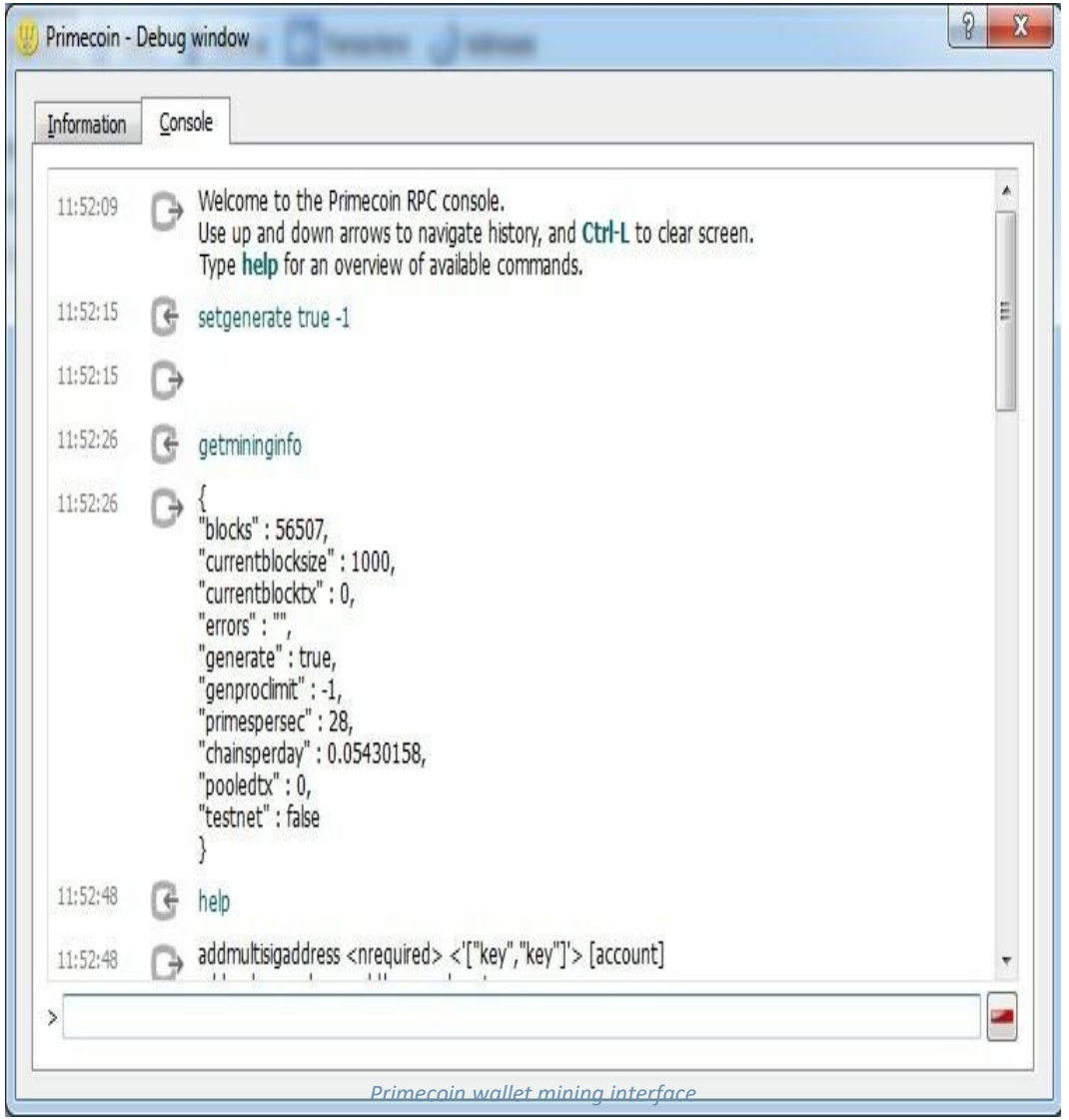
- Cunningham chains of 1st kind
- Cunningham chains of 2nd kind
- Bi-twin chains

Why prime numbers?

- Significant in physics and mathematics research
- Mining generates scientifically valuable results

Technical features:

- Difficulty adjusted every block (smoother than Bitcoin)
- No fixed coin supply limit (community-driven)
- *Market cap ~$17M (March 2018); still active*



Primecoin wallet mining interface

Zcash — Privacy Through ZK-SNARKs

Launched October 28, 2016

ZK-SNARKs: Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge

"Prove you know something without revealing what you know"

ZK-SNARK Properties:

- Complete — prover can always convince verifier of truth
- Sound — false statements cannot be proven
- Succinct — tiny proof messages
- Non-interactive — no back-and-forth needed

Address Types:

- Z-address (shielded) — fully private via ZKP
- T-address (transparent) — like Bitcoin addresses

Attribute	Value
Name	Zcash
Launch date	28/10/16
Main purpose	Currency
Currency Code	ZEC
Maximum coins	21 million
Block time	10 minutes
Consensus facilitation algorithm	Proof of Work (equihash)
Difficulty adjustment algorithm	DigiShield V3 (modified)
Mining hardware	CPU, GPU
Difficulty adjustment period	1 block

Zcash attributes summary (textbook)

Zcash: Technical Architecture

Privacy Architecture

DAP Scheme

Decentralized Anonymous Payment — transactions reveal NO information about:

- Sender address
- Receiver address
- Transaction amount

Equihash PoW

- Memory-hard, ASIC-resistant
- Based on Generalized Birthday Problem
- Very efficient verification
- DigiShield v3.0 for difficulty adjustment

Setup & Mining

Toxic Waste Ceremony

ZK-SNARKs require public parameters (proving + verifying keys). Random numbers used must be destroyed after setup.

- Multi-party computation from independent locations
- All participants must be compromised to break security
- Equipment physically destroyed after ceremony

Initial Slow Mining

- Block reward increases gradually over first 20,000 blocks
- Allows network scaling and early testing
- First-day price reached ~25,000 USD due to scarcity

Creating an Altcoin: Key Parameters

When forking Bitcoin's code, developers must configure these parameters to create a distinct new cryptocurrency:

Parameter	Description	Bitcoin Value
Consensus Algorithm	PoW, PoS, PoC, or hybrid	SHA-256 PoW
Hashing Algorithm	SHA-256, Scrypt, X11, Equihash, etc.	SHA-256
Inter-block Time	Target time between blocks	10 minutes
Block Reward	Coins awarded per mined block	12.5 BTC
Reward Halving	When block reward halves	Every 4 years
Total Coin Supply	Maximum coins ever generated	21 million BTC
Difficulty Algorithm	How difficulty responds to hash rate changes	Every 2,016 blocks

Initial Coin Offerings (ICOs)

What Are ICOs?

What is an ICO?

Fundraising for blockchain projects using cryptocurrency
Similar to an IPO but for decentralized applications

How it works:

1. Project issues new tokens
2. Investors buy tokens with BTC/ETH or fiat
3. Token value rises as project gains users

Notable ICOs:

- Ethereum (2014) — raised $18M
- Tezos — $232M
- Filecoin — $250M+

ICO vs. IPO & ERC20

ICO vs. IPO

IPO: regulated, shares in company, fiat currency
ICO: largely unregulated, tokens, cryptocurrency

ERC20 Standard

Ethereum token interface standard (EIP-20)
Defines: totalSupply, balanceOf, transfer, approve, allowance
Any Ethereum wallet supports any ERC20 token
Made token launches much easier

Risks:

- Scam ICOs and pump-and-dump schemes
- SEC may classify tokens as securities
- KYC/AML requirements under development

Key Terms & Definitions

Altcoin

Any cryptocurrency that is not Bitcoin

Hard Fork

Permanent blockchain split creating two separate chains

Merged Mining

Mining two blockchains simultaneously with one computation

Proof of Stake (PoS)

"Virtual mining" — rights proportional to coin ownership

ZK-SNARK

Prove knowledge of something without revealing that knowledge

Scrypt

Memory-hard PoW hash function originally used in Litecoin

Colored Coin

Bitcoin with metadata attached representing a digital asset

ICO

Cryptocurrency-based crowdfunding for blockchain projects

Benefits vs. Limitations of Altcoins

✓ Benefits

- PoS reduces energy consumption ~99% vs PoW
- Zcash: truly anonymous transactions via ZK-SNARKs
- Litecoin: 4× faster block time than Bitcoin
- Namecoin: first decentralized DNS system
- Primecoin PoW generates scientific prime numbers
- ICOs enable rapid capital formation for projects
- Colored coins / ERC20 tokens enable programmable assets

⚠ Limitations

- Many altcoins are pump-and-dump scams
- PoS: "nothing at stake" vulnerability
- ZK-SNARK: requires trusted setup ceremony
- ASIC resistance eventually overcome (Scrypt, X11)
- Fast block times can destabilize the chain
- ICOs are largely unregulated → fraud risk
- Fragmented ecosystem reduces network effect

Notable Altcoins at a Glance

Altcoin	Year	Key Innovation	PoW / Consensus
Namecoin (NMC)	2011	Decentralized DNS, merged mining, Zooko's Triangle	SHA-256 (merged)
Litecoin (LTC)	2011	Faster blocks (2.5 min), Scrypt memory-hard PoW	Scrypt
Primecoin (XPM)	2013	Proof of Useful Work: finding Cunningham prime chains	Prime chains
Zcash (ZEC)	2016	ZK-SNARKs for full transaction privacy (DAP scheme)	Equihash
Dash	2014	Dark Gravity Wave difficulty; Darksend mixing (CoinJoin)	X11
Monero (XMR)	2014	Ring signatures for inherent transaction anonymity	RandomX

Key Takeaways

1. Altcoins address Bitcoin's core limitations: privacy, energy waste, speed, and ASIC centralization
2. Proof of Stake is the most prominent PoW alternative — far lower energy but introduces "nothing at stake" risk
3. Difficulty adjustment algorithms (KGW, DGW, DigiShield) protect altcoins from pool hopping attacks
4. Namecoin pioneered merged mining; Litecoin introduced Scrypt; Zcash introduced ZK-SNARKs for privacy
5. ZK-SNARKs enable provable privacy — a prover demonstrates knowledge without revealing the actual data
6. ICOs became a major fundraising tool but carry significant fraud risk and face growing regulation
7. Creating an altcoin is technically easy — building a community and legitimate use case is the real challenge

Discussion & Q&A

Think. Reflect. Apply.

"If Bitcoin already exists and works, why do we need altcoins?"

"Would you trust a fully anonymous payment system like Zcash? What are the trade-offs?"

"If you were launching a blockchain project today, would you use an ICO to raise funds? Why?"

"Which consensus mechanism do you think is most sustainable for the future?"

References & Further Reading

Primary Source

- Imran Bashir. Mastering Blockchain, 2nd Edition. Packt Publishing, 2018.
  Chapter 8: Alternative Coins (pp. 334–400)

Key Online Resources

- CoinMarketCap — https://coinmarketcap.com/
- Bitcoin Energy Index — https://digiconomist.net/bitcoin-energy-consumption
- Zcash Official — https://z.cash
- Namecoin — https://namecoin.org
- Litecoin — https://litecoin.org
- Primecoin — http://primecoin.io
- ERC20 Standard — https://eips.ethereum.org/EIPS/eip-20

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